

SYSTEM REQUIREMENTS & FUNCTIONAL SPECIFICATION

Learn Ancient Greek with Xenophon

A role-based, accessible digital curriculum for beginning Ancient Greek

Field	Details
Document status	Draft for review
Version	0.4
Prepared for	GREK 399 Independent Study, Fall 2026
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Review cycle	Iterative; requirements may change after faculty review and usability testing

Purpose. This document defines what the site is intended to do, how users move through it, how the curriculum is organized, and what conditions must be satisfied for the independent-study implementation to be considered successful.

Executive Summary

Learn Ancient Greek with Xenophon is a browser-based course platform that combines cumulative grammar instruction, intentional lesson vocabulary, guided Greek readings, pronunciation audio, interactive practice, cultural context, progress tracking, and role-specific course management. Xenophon's works provide the central reading voice; the course uses four virtue-centered modules to connect language learning with Greek ethics, society, leadership, household life, and civic thought.

The system is implemented as a static HTML/CSS/JavaScript application with Netlify Functions and a relational database for authentication, course content, vocabulary, lesson progress, grades, and instructor views. This specification records both the current implementation and the intended end state. It deliberately labels planned capabilities so that unfinished ideas are not mistaken for deployed features.

How to Read This Draft

- **Implemented:** Present in the current code or content path and suitable for verification.
- **Partially implemented:** A working foundation exists, but authoring, validation, or user experience work remains.
- **Planned:** Defined as a requirement or curriculum commitment but not yet complete.
- **Future:** Outside the Initial MVP (Minimum Viable Product) unless time permits.

These exact labels appear in the Module I lesson-status table and the Appendix A requirements register. Section 15.1 groups Future items. Narrative sections may describe several related requirements with different statuses, so the tables are the authoritative status locations.

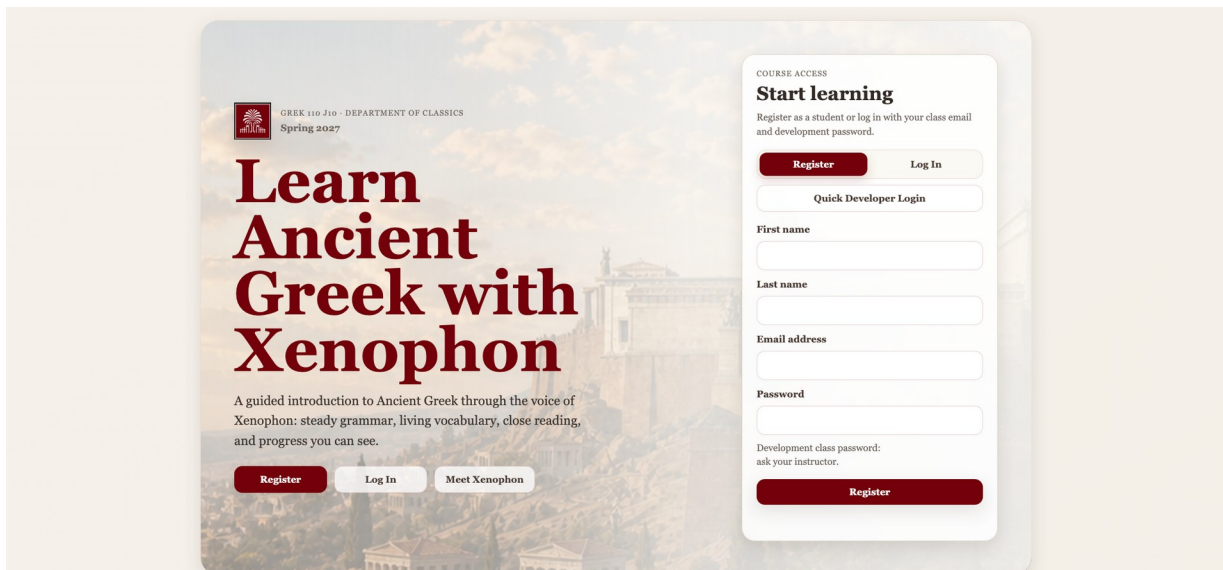


Figure 1. Current public landing and course-access screen.

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Drafting principle. Course-facing language should remain understandable to students and faculty. Technical details appear only where they define behavior, maintainability, security, or acceptance criteria.

1. Product Vision and Objectives

1.1 Product Vision

The platform will help beginning students move from the Greek alphabet to sustained reading by making vocabulary, morphology, syntax, pronunciation, and cultural interpretation parts of one coherent learning path. It should feel like a guided course rather than a collection of unrelated reference pages.

1.2 Educational Objectives

- Build reliable decoding and pronunciation habits before requiring substantial reading.
- Introduce grammar cumulatively and immediately apply each new form in meaningful Greek.
- Use Xenophon's prose and themes to connect language with history, ethics, leadership, household management, education, and civic life.
- Separate required lesson vocabulary from temporary reading glosses so mastery expectations remain clear.
- Provide frequent practice, immediate feedback, and visible progress without allowing gamification to replace learning.
- Support students using keyboards, touch devices, screen readers, magnification, and audio-based study methods.

1.3 Product Goals

Goal	Required outcome
Structured learning path	Unit 0 and four modules present lessons in a deliberate sequence with clear prerequisites and next steps.
Integrated lesson design	Every completed lesson connects vocabulary, reading, grammar, practice, cultural context, and assessment.
Role-aware operation	Students learn, professors monitor and guide, and administrators manage course content and users.
Accessible participation	Core course functions are usable without a mouse and meet the defined WCAG target.
Maintainable content	Shared renderers and structured data prevent curriculum pages from becoming isolated one-off implementations.
Verifiable progress	Completion, grades, activity, vocabulary study, and achievements are recorded and displayed consistently.

2. Scope, Assumptions, and Status

2.1 In Scope for the Course Platform

- Public course introduction, account registration, login, and multi-role view selection.
- Unit 0 for alphabet, pronunciation, breathings, accents, keyboarding, and early decoding.
- Four thematic modules with twelve lessons per module.
- Three-page lesson flow with readings, language study, cultural context, review, and quiz functions.

- Lesson vocabulary, lesson-specific reading glosses, dictionaries, flashcards, forms, principal parts, maps, and pronunciation audio.
- Student progress, grades, XP, levels, achievements, weekly goals, and recent activity.
- Professor roster/progress views and administrator content authoring/publishing workflows.
- Responsive and accessible use on contemporary desktop, tablet, and mobile browsers.

2.2 Outside the Initial MVP (Planned for Later Development)

- Open-ended AI tutoring or automated scholarly interpretation.
- A native iOS or Android application; the web application remains the primary product.
- A full learning-management-system replacement with institutional enrollment, billing, or transcript processing.
- Public community discussion, peer messaging, or social-network features without a separate moderation and privacy design.
- Broad coverage of authors beyond Xenophon and carefully selected comparative passages.

2.3 Operating Assumptions

Assumption	Specification consequence
Course identity	The current interface is branded for USC Classics and a GREK 110 course instance; course metadata should remain configurable.
Browser delivery	No installation is required beyond a standards-compliant browser; the site may be installed as a progressive web app where supported.
Iterative content	Lesson structure may stabilize before all lesson prose, exercises, and audio are complete.
Authoritative data	Database-backed published lesson content is authoritative when present; verified local fallback content supports development.
Open build access	Temporary development settings that expose unfinished lessons must be disabled or replaced by explicit release rules before production use.

3. Users, Roles, and Permissions

A user account may have one or more roles. After login, a multi-role user selects the dashboard appropriate to the task. The active role governs navigation, displayed data, and permitted operations; possessing an administrator role must not automatically expose administrative controls while the user is in Student view.

Capability	Student	Professor	Administrator
Open released lessons and resources	Yes	Yes	Yes
Complete exercises and quizzes	Yes	Yes, in learner/preview view	Yes, in learner/preview view
View own progress and grades	Yes	Yes	Yes
View class roster and aggregate progress	No	Yes	Yes
Preview an individual student dashboard	No	Yes	Yes
Review submissions and students needing attention	No	Yes	Yes
Create, edit, draft, and publish lesson content	No	No unless separately authorized	Yes

Capability	Student	Professor	Administrator
Manage users and roles	No	No	Yes
Manage course/site settings	No	Limited course preferences	Yes

3.1 Student Role

- Resume at the current lesson or select any released lesson allowed by the course sequence.
- Study lesson vocabulary, guided readings, grammar, culture, and review content.
- Complete exercises, quizzes, flashcards, principal-parts practice, and other assigned activities.
- Use the on-screen polytonic keyboard whenever a Greek response is required.
- View personal profile, progress, grades, XP, achievements, weekly goals, and recent activity.

3.2 Professor Role

- View a live roster with each student's progress, current lesson, level, average grade, recent activity, and status.
- Identify students needing attention based on inactivity or grade thresholds.
- Review submissions and grade distributions.
- Open a read-only student-dashboard preview and return to the professor dashboard without changing the professor's role or the student's data.
- Use course content and resources in the same way as a learner when preparing instruction.

3.3 Administrator Role

- Manage users, roles, course settings, lessons, exercises, grades, and site configuration.
- Create lesson drafts, validate structured content, and publish approved versions.
- Maintain lesson vocabulary, reading glosses, audio references, banners, citations, and assessment content.
- Preview content in student-facing form before publishing.
- Retain revision notes and prevent unvalidated content from replacing the published lesson.

4. Information Architecture and Side Navigation

Authenticated pages use a persistent left navigation panel. The panel combines institutional/course identity, role-appropriate navigation, a collapse control, and a profile area. It is the application's main orientation mechanism and must behave consistently across dashboards, modules, lessons, maps, and reference resources.



Figure 2. Current Module I page with the persistent side navigation.

4.1 Side Navigation Requirements

- NAV-01. The panel shall display course identity and term consistently on authenticated pages.
- NAV-02. Navigation items shall be generated from the active role configuration rather than duplicated independently on each page.
- NAV-03. The current page shall be visually distinguished; exact-page links shall expose aria-current=page.
- NAV-04. A module or lesson page shall keep Lessons highlighted as the active section; resource subpages shall keep Resources highlighted.
- NAV-05. The collapse/expand button shall expose its state programmatically and retain a readable label and keyboard focus.
- NAV-06. On narrow screens, navigation shall not cover essential content or trap keyboard focus.
- NAV-07. Grouped reference links such as FORMS shall preserve visible group-heading semantics rather than appearing as unrelated flat links.
- NAV-08. Unimplemented destinations shall not appear as working links in a production release; they shall be hidden, labeled as planned, or routed to an explanatory page.

4.2 Role-Specific Primary Navigation

Role	Primary destinations
Student	Dashboard; Introduction; Lessons; Flashcards; Maps; Resources; Profile; Feedback
Professor	Dashboard; Students; Progress; Submissions; Introduction; Lessons; Maps; Resources; Feedback; Discussions; Settings
Administrator	Dashboard; Users & Roles; Introduction; Lessons; Maps; Resources; Feedback; Exercises; Gradebook; Course Content; Site Settings



Figure 3. Dashboard visual reference showing navigation, progress, learning path, recent activity, and achievements.

Figure 3 is a design reference retained in the repository. It guides hierarchy and information density; it is not a promise of pixel-identical final rendering.

5. Curriculum Architecture

The course follows a Unit → Module → Lesson hierarchy. Unit 0 establishes reading readiness. Four subsequent modules organize grammar and reading around Greek virtue concepts. Each module contains twelve lessons, producing a planned forty-eight-lesson course after Unit 0.

Sequence	Theme	Learning emphasis	Lessons
Unit 0	Greek Alphabet & Reading Readiness	Letters, sounds, diphthongs, breathings, accents, keyboarding, early decoding	1 introductory unit
Module I	σοφία — Wisdom and Socrates	Learning, inquiry, self-knowledge, and the examined life	Lessons 1–12
Module II	ἀνδρεία — Courage and Leadership	Endurance, fear, leadership, future action, aorist narrative	Lessons 13–24
Module III	σωφροσύνη — Self-Control and Discipline	Self-mastery, household order, middle voice, purpose, conditionals	Lessons 25–36
Module IV	δικαιοσύνη — Justice and the City	Law, responsibility, reported speech, clauses, argument	Lessons 37–48



Figure 4. Visual identity and thematic progression of the four course modules.

5.1 Standard Lesson Anatomy

- Overview and learning objectives
- Intentional lesson vocabulary with dictionary information and available pronunciation audio
- Guided reading with paragraph-level glosses and source citation
- Word study and grammar explanation
- Practice exercises with immediate feedback

- Greek-world cultural or philosophical context
- Review, lesson test, completion state, and next-step navigation

Content boundary. Lesson Vocabulary defines words students are intentionally expected to study and may feed global dictionaries, flashcards, review, and quizzes. Reading Glosses are temporary passage-specific aids and shall remain inside the lesson reading unless an editor explicitly promotes the word into Lesson Vocabulary.

6. Module I Specification

6.1 Module Purpose

Module I uses σοφία as both theme and learning method. Students meet Xenophon first through home, household, education, preparation, and Socratic inquiry. The language sequence begins with essential sentence recognition and gradually adds cases, agreement, tense, infinitives, participles, and more sustained reading. Primary literary orientation comes from Xenophon's Memorabilia and related Socratic works, with other Xenophontic settings used where pedagogically useful.

Lesson	Title	Primary language focus	Status
1	Xenophon at Home	Nominative and accusative singular; present active indicative; definite article; noun/adjective agreement	Partially implemented
2	The Household of Xenophon	Second-declension nouns; adjective agreement; possessive genitives; εἰμί; simple prepositions	Partially implemented
3	The Education of Xenophon	Third-person present verbs; direct objects; infinitives; introductory middle/passive; demonstratives	Partially implemented
4	The Preparation Before the March	Singular noun cases; singular adjective agreement; possessive genitive; natural case usage	Partially implemented
5	Learning Through Questioning	Adjectives and agreement; attributive and predicate position	Planned
6	The Search for Knowledge	Second-declension nouns; prepositions with cases	Planned
7	Examining Oneself	Present middle/passive voice; reflexive sense	Planned
8	In the Agora	Expanded prepositions; introduction to the dative	Planned
9	Socrates Questions All	Imperfect tense; continuous and repeated past action	Planned
10	To Know and To Learn	Introductory and complementary infinitives	Planned
11	The Thinking Mind	Present active participle; introductory participial description	Planned
12	The Examined Life	Module review: present, imperfect, infinitives, participles	Planned

6.2 Detailed Requirements for Lessons 1-4

Lesson 1

Students identify basic subjects, objects, articles, present-tense verbs, and agreement in a home-centered reading. Practice covers verb meaning and recognition, noun cases, adjective agreement, and the definite article.

Lesson 2

Students expand the nominal system through household vocabulary, second-declension forms, possessive relationships, εἰμί, and simple spatial relationships. The reading connects language to Xenophon's household world.

Lesson 3

Students read about education and training while adding third-person verbs, accusative objects, infinitives, demonstratives, and an introductory view of middle/passive morphology.

Lesson 4

Students consolidate singular case usage and adjective agreement in a preparation-before-the-march narrative. The lesson shall include its published banner, three reading paragraphs, and paragraph-specific glosses.

6.3 Authoring Standard for Lessons 5–12

- Each lesson shall define measurable language objectives, a reading purpose, a cultural/philosophical connection, and a completion test.
- New vocabulary shall be limited to an intentional learnable set; incidental reading words shall use passage glosses.
- Exercises shall include recognition and recall, and at least one task requiring the student to interpret or produce Greek.
- Grammar explanations shall build on prior forms and explicitly identify prerequisites.
- Published readings shall include a human-readable source citation and editorial status indicating adapted, lightly adapted, or authentic text.
- Audio placeholders shall not be mistaken for finished audio; release status shall distinguish available, pending, and intentionally omitted audio.

7. Lesson Experience and Functional Behavior

7.1 Three-Page Lesson Flow

Page	Purpose	Required components
Page 1 — Reading	Orient the student and establish vocabulary and passage context.	Banner; overview; objectives; lesson vocabulary; guided reading; audio status; paragraph glosses; reading tools
Page 2 — Language Study	Explain and practice the forms needed for the reading.	Word building; grammar sections; examples; forms links; practice gates; immediate feedback; keyboard where Greek entry is required
Page 3 — Greek World / Review / Quiz	Connect language to culture, consolidate learning, and record achievement.	Cultural feature; interpretive questions; grammar summary; review; lesson test; score; completion; next lesson

7.2 Core Lesson Requirements

- LES-01. The lesson header shall show lesson number, English title, optional Greek title, module identity, and an informative banner image or accessible placeholder.
- LES-02. Objectives shall describe observable learner outcomes rather than only list topics.
- LES-03. Vocabulary entries shall support lemma, display form, gloss, part of speech, dictionary form, morphology metadata, and audio when available.

- LES-04. Reading paragraphs shall preserve Greek text, optional translation, notes, source citation, and paragraph-level glosses.
- LES-05. Glosses shall expose Greek form and English aid and may include lemma and morphology; they shall not silently enter global study lists.
- LES-06. Exercise feedback shall be immediate, specific, and announced to assistive technology when it changes dynamically.
- LES-07. A lesson shall not be marked complete until the defined completion condition is satisfied and persisted.
- LES-08. Previous/next navigation shall identify the destination and preserve the student's place in the lesson sequence.
- LES-09. Administrators shall be able to preview a draft without overwriting the published version.
- LES-10. When the lesson API is unavailable during development, the interface may use a verified fallback, but production shall surface a recoverable error rather than silently showing incorrect content.

7.3 Core Learning Loop

1. Learn the intentional vocabulary.
2. Hear and pronounce the words and passage where audio is available.
3. Read the passage with limited, lesson-specific gloss support.
4. Study the grammar needed to understand the passage.
5. Practice recognition, recall, parsing, and translation with immediate feedback.
6. Connect the language to Xenophon's cultural or philosophical setting.
7. Complete review and assessment, then revisit weak areas.

8. Learning Tools and Reference Resources

8.1 Polytonic Greek Keyboard

The built-in keyboard removes the requirement that students install an operating-system Greek layout before they can participate. It is a reusable QWERTY-style component activated for designated Greek input fields. Students may click on-screen keys or use the matching physical-key shortcuts.

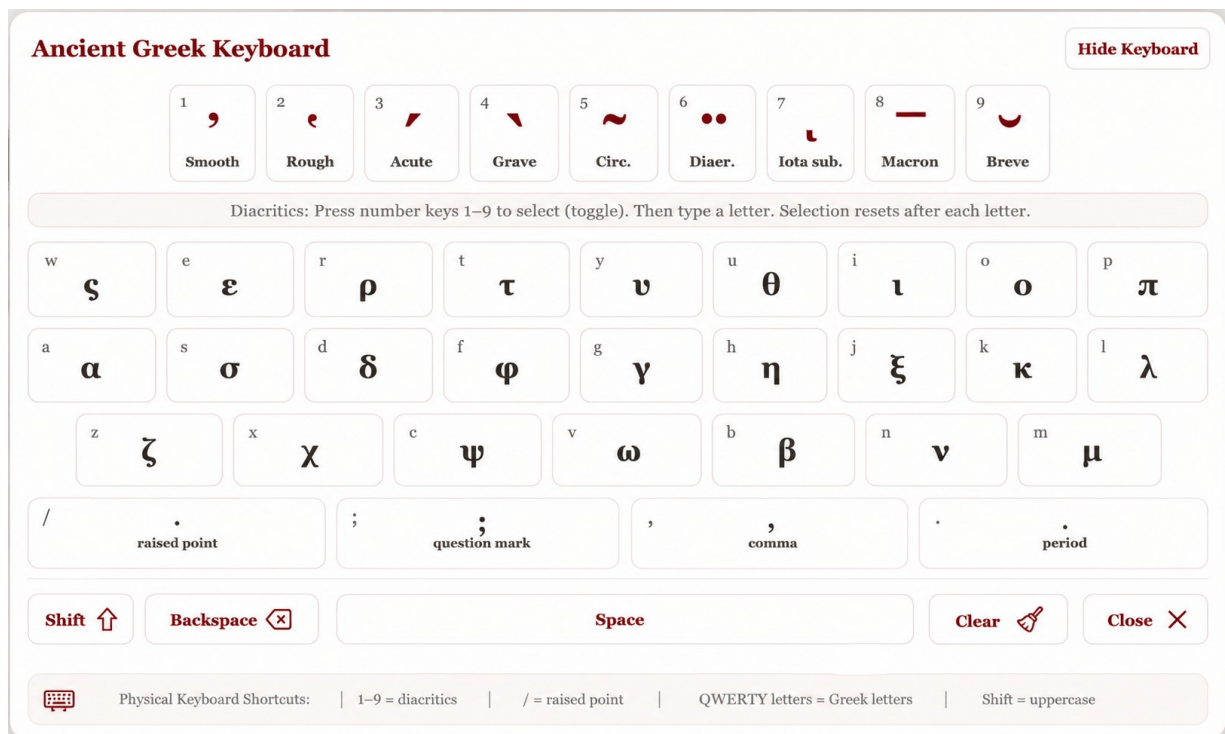


Figure 5. Polytonic Ancient Greek keyboard layout reference.

- KEY-01. QWERTY letter keys shall map consistently to Greek letters, including final sigma behavior where supported.
- KEY-02. Diacritic controls shall include smooth and rough breathing, acute, grave, circumflex, diaeresis, iota subscript, macron, and breve.
- KEY-03. Conflicting marks shall be normalized so impossible combinations are not retained.
- KEY-04. Composed output shall use normalized Unicode rather than visually similar custom glyphs.
- KEY-05. The keyboard shall support uppercase, backspace, space, clear, Greek punctuation, close/hide, and physical-key shortcuts.
- KEY-06. Every key shall have an accessible name, selected diacritics shall expose state, and pending composition changes shall be announced.
- KEY-07. Opening, closing, or dragging the keyboard shall not trap focus or block the target field on small screens.

8.2 Vocabulary and Dictionaries

- Greek → English and English → Greek dictionaries shall be built only from intentional Lesson Vocabulary entries.
- Dictionary results shall preserve useful lexical metadata such as lemma, principal parts, gender, genitive, adjective forms, definitions, and articles where applicable.
- Global flashcards, vocabulary review, and vocabulary quizzes shall use the same intentional vocabulary boundary.
- Reading glosses shall remain lesson-specific aids and shall not inflate vocabulary mastery counts.

8.3 Forms and Principal Parts

Forms pages provide reusable paradigms for the definite article, first-, second-, and third-declension nouns, adjectives, pronouns, omega verbs, participles, and εἶμι. A shared data source and renderer shall generate both the Forms landing entries and individual paradigm pages. Principal Parts is a related resource with a shared verb dataset, study views, randomized practice, progress credit, and achievement hooks.

8.4 Audio

- Support word-level pronunciation and passage-level listening.
- Provide an accessible text equivalent or transcript for instructional audio.
- Identify unavailable audio honestly rather than rendering a nonfunctional control.
- Use a consistent documented pronunciation model across generated and recorded assets.
- Allow future slow playback without changing the underlying authoritative text.

8.5 Maps and Cultural Resources

Maps support geographic orientation for Athens, the Greek world, Xenophon's travels, and reading-specific locations. Cultural resources should be linked from lessons when they clarify a passage, not added as disconnected enrichment. Maps require text alternatives or equivalent descriptions for information that cannot be perceived visually.

9. Progress, Assessment, and Motivation

Area	Required behavior
Lesson progress	Record not started, in progress, and completed states; show current location and next recommended step.
Assessment	Record lesson-test attempts and best/defined scores; distinguish practice from graded assessment.
Vocabulary	Track only intentional Lesson Vocabulary mastery; do not count passage glosses.
Activity	Record meaningful events such as lesson completion, practice, reading, audio, and assessment.
XP and levels	Award documented values consistently and prevent duplicate credit for one-time actions.
Achievements	Compute earned and locked achievements from explicit criteria and provide meaningful names and descriptions.
Weekly goal	Show a configurable target and progress toward it without penalizing inaccessible or unavailable activities.
Professor analytics	Aggregate progress and grades without exposing one student's data to another student.

10. Professor and Administrator Workflows

10.1 Professor Dashboard

- Load the current course roster and student-progress summaries from the server-side course data path.

- Display class overview, submissions/grading, weekly activity, student progress, attention indicators, lesson progress, and grade distribution.
- Provide a horizontally and vertically scrollable progress table that remains keyboard accessible.
- Allow a professor to preview a selected student dashboard without gaining permission to alter that student's recorded work.
- Handle unavailable data with a clear loading or error message rather than fabricated live results.

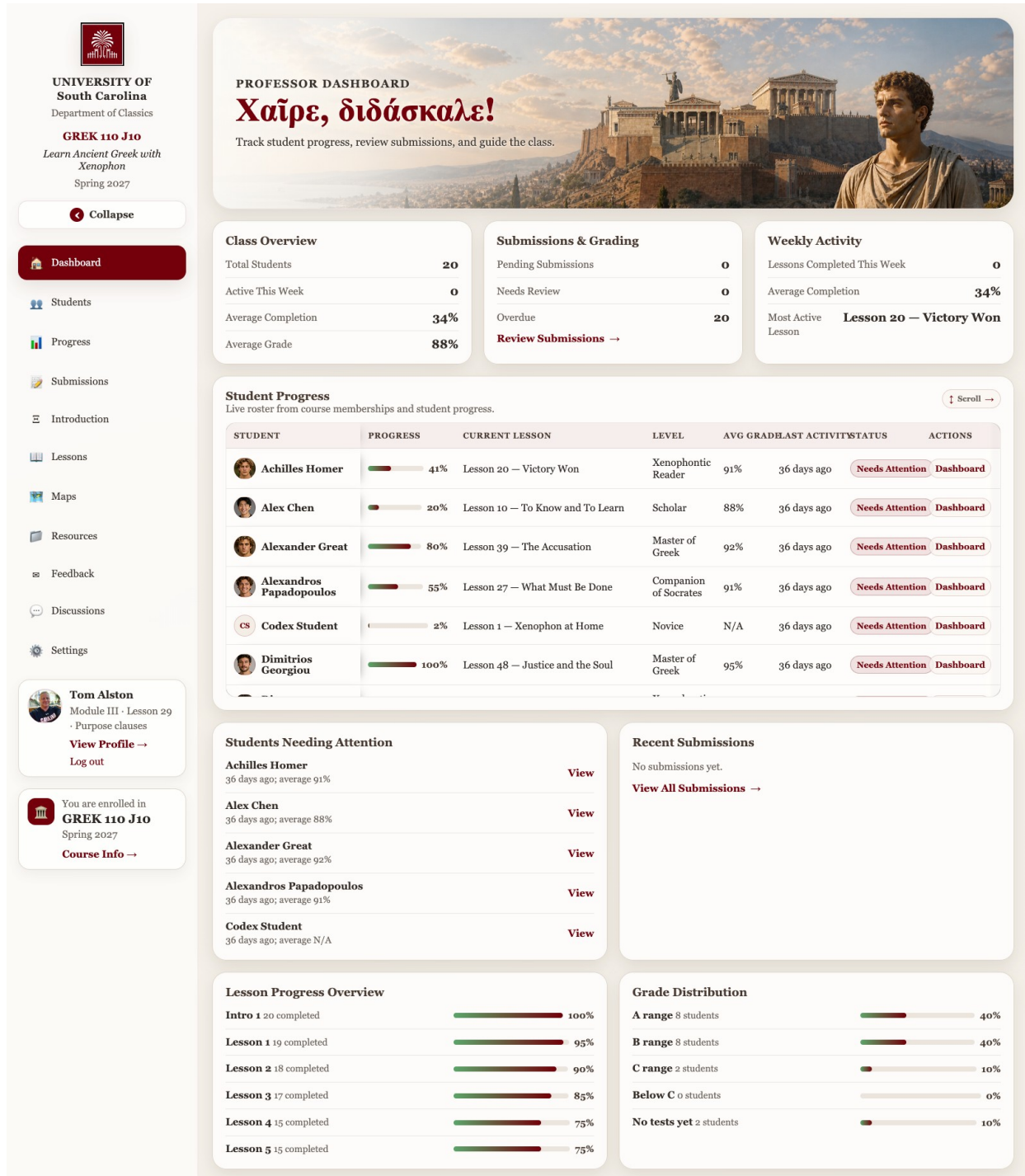


Figure 6. Current Professor Dashboard with class summary, student progress, attention indicators, submissions, lesson completion, and grade distribution.

10.2 Administrator Content Workflow

- Select a lesson and load the latest published content and any saved draft.
- Edit structured fields for titles, banners, objectives, vocabulary, readings, glosses, word study, grammar, culture, exercises, quiz, citations, and page metadata.
- Validate types and required relationships before saving or publishing.
- Save a draft with an editorial note without changing the student-facing published version.
- Preview the draft through the normal renderer.
- Publish an approved revision, recording version and update information.
- Reject unauthorized write requests even if a client manually exposes an editing control.

11. Content, Data, and Technical Architecture

11.1 Hosting and Runtime Architecture

Layer	Current implementation
Presentation	Root-level static HTML pages, shared CSS, vanilla JavaScript renderers, images, audio references, and installable-web-app metadata served from Netlify's CDN.
Shared application behavior	Role dashboards, navigation, modules, lesson listing, profile hydration, progress, achievements, and mobile learning behavior.
Lesson renderer	A shared lesson page consumes structured lesson content and renders reading, vocabulary, grammar, culture, exercises, and quiz sections.
Server functions	Twelve TypeScript Netlify Functions provide login, registration, password changes, courses, profiles, profile photos, lesson metadata/content, progress, dictionary, professor dashboard, and administrative lesson writes.
Database	A Neon-hosted PostgreSQL 17 database stores identity, enrollment, curriculum, language content, progress, grades, activity, and publishing records.
Object storage	Netlify Blobs stores uploaded profile-photo bytes; PostgreSQL stores only the resulting photo URL.
Deployment	Netlify publishes the repository root as the static site and bundles netlify/functions with esbuild. The Neon connection string is supplied to Functions through the NETLIFY_DATABASE_URL environment variable.

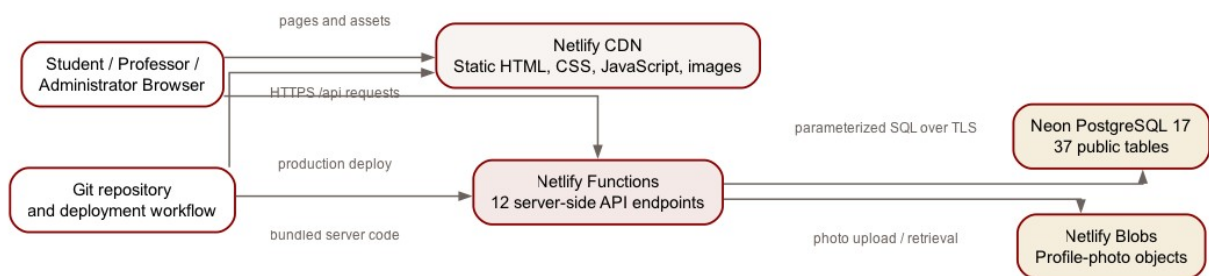


Figure 7. Current hosting, function, database, and object-storage architecture.

11.2 Request and Data Flow

- A browser first downloads static HTML, CSS, JavaScript, images, and web-app metadata from Netlify's content-delivery network.
- Operations requiring identity, protected data, or persistence call a Netlify Function through an /api endpoint.

- The Function validates the request and active role, executes parameterized PostgreSQL queries over TLS, and returns JSON or a controlled error response.
- The browser renderer converts the response into the role-appropriate page without exposing the database connection string or privileged SQL.
- Profile-photo uploads pass through a Function to Netlify Blobs; ordinary course images remain versioned static repository assets.

11.3 Database Structure

The live public schema currently contains thirty-seven base tables. The design is relational: stable entities use UUID or text primary keys, many-to-many relationships use junction tables, foreign keys enforce ownership, and JSONB is reserved for content whose editorial shape is expected to evolve. Indexes support common lesson, user, status, chronology, dictionary, and morphology lookups.

Domain	Tables	Purpose
Identity and enrollment (7)	users; user_credentials; user_profiles; roles; user_roles; courses; course_memberships	Accounts, hashed credentials, profile data, active roles, course instances, and enrollment.
Curriculum and publishing (7)	modules; lessons; lesson_segments; lesson_content_blocks; lesson_content_overrides; lesson_content_drafts; lesson_content_versions	Ordered course structure plus editable, previewable, versioned lesson content.
Language content (5)	readings; reading_glosses; vocabulary_items; lesson_vocabulary; audio_assets	Greek readings, passage-only glosses, intentional vocabulary assignments, dictionary metadata, and audio references.
Assessment and progress (9)	exercises; exercise_choices; quizzes; quiz_questions; quiz_answer_options; exercise_attempts; student_lesson_test_grades; student_progress; lesson_progress	Practice, assessment, attempts, grades, current position, completion state, XP, and weekly goals.
Engagement and study (6)	activity_events; levels; achievements; user_achievements; flashcard_reviews; flashcard_exports	Activity history, motivation, spaced-study state, and exports.
Course communication (3)	announcements; discussions; discussion_posts	Instructor communication and course/lesson discussion records.

11.4 Key Relationships and Data Boundaries

- A user has one credential record and profile, may hold multiple roles through user_roles, and may join multiple course instances through course_memberships.
- A course contains ordered modules; a module contains ordered lessons; a lesson contains segments, content blocks, readings, exercises, quizzes, and progress records.
- Lessons and reusable vocabulary items have a many-to-many relationship through lesson_vocabulary. Only this intentional lesson vocabulary may feed global dictionaries, flashcards, review, and vocabulary quizzes.
- reading_glosses belongs to a lesson and optionally a reading. These rows are temporary passage aids and do not become global study vocabulary without an explicit editorial promotion.
- lesson_content_drafts stores private work in progress; lesson_content_overrides stores the published structured lesson snapshot; lesson_content_versions preserves recoverable history.
- Per-student operational growth is concentrated in lesson_progress, exercise_attempts, student_lesson_test_grades, activity_events, flashcard_reviews, and user_achievements.

11.5 Content Source of Truth

Published database content is the preferred source for lesson delivery. The lesson-content function assembles lesson metadata, pages, intentional vocabulary, reading text, reading glosses, and published content blocks. Development fallback content may protect local work from temporary service failures, but verification must inspect the actual payload delivered to the lesson renderer.

Verification rule. When content or an asset does not appear in the interface, testing must follow the real serving path—database or migration, Netlify Function response, renderer input, and final page—not stop after confirming that a source file contains the expected text.

11.6 Current Database Snapshot

Observed item	Current value	Interpretation
Database engine	PostgreSQL 17.10 on Neon	Current connected production/development database provider and engine.
Database size	13 MB	Small relative to even the current Neon Free-plan storage allowance; fixed schema/index overhead is a material portion.
Public tables	37	Matches the migration-defined schema and live verification output.
Accounts / memberships	22 / 22	Current seeded and working classroom-scale population.
Lessons / vocabulary items	49 / 258	Unit 0 plus the forty-eight-lesson course structure and current reusable vocabulary records.
Lesson progress / test grades	1,030 / 301 rows	Representative per-user progress and assessment growth.
Direct connection ceiling	901 total; 7 Neon-reserved	Database setting reported by SHOW max_connections-equivalent inspection; it is not a promise of 894 simultaneous students.
Connection mode	Direct, not pooled	Acceptable for development and modest traffic, but pooled connections are recommended before sustained serverless concurrency.

Snapshot captured August 5, 2026 through read-only schema, row-count, size, version, and connection-setting queries against the configured Neon database.

11.7 Provider Limits and User Capacity

Capacity conclusion. The current platform is comfortably sized for a normal university class and should be planned initially for 25–50 simultaneously active students. Several hundred registered users with staggered activity are a reasonable design target, but this is not a certified maximum. A public claim above classroom scale requires load testing and confirmation of the actual Neon and Netlify billing plans.

Limit or factor	Current or published value	Effect on this application
Neon direct connections	901 total for the current endpoint; seven are reserved	Connection count is unlikely to constrain one class, but direct per-invocation connections can spike under serverless bursts.
Neon pooled connections	Up to 10,000 client connections through PgBouncer	Use a pooled Neon hostname before broad deployment; transactions still share a smaller set of active database connections.
Neon Free storage, if applicable	0.5 GB per project	The current 13 MB database uses only a small fraction, but uploaded media must remain outside PostgreSQL and table growth must be monitored.
Neon Free compute, if applicable	100 CU-hours per project per month; up to 2 CU / 8 GB RAM; scale to zero after inactivity	Intermittent class use is favored. Sustained all-day activity or heavy analytics can exhaust compute allowance before storage.
Neon Free network transfer, if applicable	5 GB public transfer per month	Keep function responses compact, paginate professor views, and avoid returning unused content.
Netlify credit plan, if applicable	Free plan: 300 monthly credits; requests, bandwidth, compute, and production deploys share the balance	Static assets, API calls, Function execution, and deploy frequency all consume the hosting allowance. Legacy Netlify plans use different limits.

Provider references checked August 5, 2026: <https://neon.com/pricing>; <https://neon.com/docs/connect/connection-pooling>; <https://neon.com/docs/manage/endpoints/>; <https://www.netlify.com/pricing/>; and <https://docs.netlify.com/manage/accounts-and-billing/billing/billing-for-credit-based-plans/how-credits-work/>.

11.8 Workload Planning and Validation

Usage level	Planning assessment	Required action
One class: 25–50 active at once	Expected to be suitable for the current application and database size.	Run a scripted lesson/login/progress test before the semester and watch Function errors, latency, database compute, and connection graphs.
Several classes: about 100 active at once	Likely feasible, but not yet demonstrated.	Switch to the pooled Neon connection, eliminate N+1 queries, paginate dashboards, and execute a 100-user load test.
Large program: 250+ active at once	Outside the current verified operating envelope.	Benchmark representative reads and writes, review paid provider tiers, configure monitoring/alerts, and establish performance objectives.
Registered but mostly inactive users	Hundreds should require little storage; thousands may fit technically, but monthly compute, requests, and transfer become the controlling quotas.	Retain/archive policy, usage monitoring, and periodic storage/index review are required.

- Performance test scenarios shall include login, lesson retrieval, dictionary search, progress writes, quiz/test submission, professor dashboard loading, and administrator save/preview/publish.
- Tests shall record p50 and p95 response latency, error rate, Function duration, database CPU/compute usage, open and pooled connections, rows returned, and transfer volume.
- The release target shall be revised from an estimate to a measured capacity only after a repeatable load test passes without data loss, authorization failure, or sustained quota exhaustion.

11.9 Security and Authorization

- Server-side functions shall validate authorization for professor and administrator operations; client-side role visibility is not a security boundary.
- Passwords shall not be stored in browser-visible plaintext in production.
- Database credentials and service keys shall remain in environment-managed secrets and never be embedded in static JavaScript.
- User-facing error messages shall avoid exposing queries, stack traces, connection strings, or internal identifiers.
- Administrative publication shall validate structured content and preserve recoverable draft/published history.

12. Accessibility and Inclusive Design

Recommended target. The original project goal names WCAG 2.1. For new work in 2026, this specification adopts WCAG 2.2 Level AA as the target because W3C encourages use of the latest WCAG 2 version and WCAG 2.2 is designed to remain backward compatible with 2.1.

12.1 Accessibility Requirements

- A11Y-01. All core learning and administrative workflows shall be operable by keyboard alone.
- A11Y-02. Focus order shall follow the visual/reading order; focus indicators shall be visible and not obscured by sticky or floating controls.
- A11Y-03. Pages shall use semantic headings, landmarks, navigation labels, lists, tables, and form associations.
- A11Y-04. Greek passages and isolated Greek terms shall use an appropriate language declaration such as lang=grc; the surrounding interface shall retain its document language.
- A11Y-05. Informative images shall have concise alternatives; decorative images shall be hidden from assistive technology.
- A11Y-06. Text, controls, focus indicators, and status badges shall meet Level AA contrast requirements in default and interactive states.
- A11Y-07. Text shall remain usable at 200% zoom and during browser text resizing without horizontal scrolling for ordinary reading content.
- A11Y-08. Dynamic exercise feedback, validation errors, keyboard composition state, and save/publish status shall be announced without moving focus unnecessarily.
- A11Y-09. Audio instruction shall include a transcript or equivalent text; audio controls shall have accessible names and keyboard operation.
- A11Y-10. Touch and pointer targets shall meet the WCAG 2.2 minimum target-size requirement or documented exceptions.
- A11Y-11. Dragging shall not be the only way to move or use the on-screen keyboard; equivalent controls shall exist.
- A11Y-12. Authentication shall not require a cognitive-function test and shall work with password managers and appropriate autocomplete tokens.
- A11Y-13. Responsive layouts shall preserve content order, labels, and controls on narrow screens.
- A11Y-14. Automated checks and manual keyboard/screen-reader testing shall be part of release verification.

12.2 Accessibility Acceptance Test Set

Test	Pass condition
Keyboard path	A user can register/log in, select a role, navigate, complete a lesson activity, use the Greek keyboard, submit an answer, and log out without a mouse.
Focus visibility	Focused controls remain visible, including when the sidebar or Greek keyboard is open.
Screen reader structure	Landmarks, headings, labels, table headers, selected states, errors, and dynamic feedback are announced meaningfully.
Zoom/reflow	At 200% zoom and a narrow viewport, content reflows without clipping essential text or controls.
Color/contrast	Information does not depend on color alone and all required contrast measurements pass.
Media alternative	Every required audio item has equivalent instructional text; informative screenshots and banners have meaningful alternatives.

Accessibility standard reference: W3C Web Accessibility Initiative, "WCAG 2 Overview" and Web Content Accessibility Guidelines (WCAG) 2.2, <https://www.w3.org/WAI/standards-guidelines/wcag/> and <https://www.w3.org/TR/WCAG22/>.

13. Nonfunctional Requirements

Category	Requirement
Performance	Primary pages should become usable promptly on ordinary campus/home connections; images and audio shall be optimized and loaded deliberately.
Responsive design	Core screens shall support contemporary desktop, tablet, and mobile widths without separate content versions.
Browser support	Current versions of Chrome, Safari, Firefox, and Edge shall support all core functions; progressive enhancement shall protect basic reading.
Reliability	Network or database failures shall produce recoverable messages; user work shall not be falsely marked complete or silently discarded.
Maintainability	Shared data and renderers shall drive repeated structures such as modules, lessons, forms, navigation, achievements, and principal parts.
Content integrity	Greek text, diacritics, translations, citations, and lexical metadata shall survive storage and rendering without normalization loss.
Privacy	Users shall see only information appropriate to their role; student analytics shall not be exposed to other students.
Observability	Server and content failures shall be diagnosable without exposing sensitive details to users.
Backup/recovery	Published content and schema changes shall be versioned or backed up so an erroneous publication can be recovered.

14. Independent-Study MVP and Acceptance Criteria

14.1 Proposed MVP

For the GREK 399 independent study, the recommended MVP is a coherent, demonstrable course foundation rather than partial implementation of all forty-eight lessons. The MVP should include the complete platform framework, Unit 0, and at least four fully authored Module I lessons, with a fifth and sixth lesson included if the semester schedule permits.

- Public landing, registration/login, role selection, and stable Student/Professor/Administrator views.
- Unit 0 alphabet and reading-readiness experience with pronunciation audio and keyboard practice.
- Module I introduction and at least Lessons 1–4 complete across all three lesson pages.
- Intentional lesson vocabulary, lesson-specific glosses, grammar, practice, cultural context, citation, and lesson test for each completed lesson.
- Functional polytonic keyboard, dictionaries, flashcards, forms, principal parts, maps, and progress display at a documented level of completeness.
- Professor progress view and an administrator draft/preview/publish workflow.
- Accessibility evaluation against the test set in Section 12 and documented remediation of critical issues.
- A final demonstration plus supporting curriculum/technical documentation and a prioritized post-semester backlog.

14.2 Acceptance Criteria

ID	Acceptance criterion
AC-01	A new user can create or access an account and reach the correct role dashboard.
AC-02	A student can move from Unit 0 through each released Module I lesson using clear previous/next navigation.
AC-03	Each MVP lesson renders the correct published vocabulary, reading, glosses, grammar, exercises, culture content, and quiz.
AC-04	Student completion and assessment results persist and appear in the student's dashboard.
AC-05	Professor view reports roster progress and supports read-only student-dashboard preview.
AC-06	Administrator can save a draft, preview it, publish it, and recover the prior published version or data backup.
AC-07	Global vocabulary tools include only intentional Lesson Vocabulary, not Reading Glosses.
AC-08	The Greek keyboard produces valid normalized polytonic text through pointer and physical-key input.
AC-09	The required keyboard, zoom/reflow, contrast, labeling, and screen-reader checks pass or have documented exceptions approved for remediation.
AC-10	The demonstration can run from the deployed environment without relying on unexplained developer-only shortcuts.

15. Future Enhancements and Open Decisions

15.1 Future Enhancements

- Complete authoring and validation of Modules II–IV.
- Adaptive review and spaced repetition based on demonstrated weakness.
- Expanded passage audio, slow playback, synchronized highlighting, and listening exercises.
- More sophisticated assignment scheduling and professor feedback tools.
- Export or integration options for institutional LMS gradebooks, subject to USC requirements.
- Additional authors and comparative passages after the Xenophon-centered curriculum is stable.
- AI-assisted parsing feedback only after scholarly guardrails, privacy review, and transparent error handling are defined.

15.2 Open Decisions for Faculty Review

Decision	Question to resolve
Pronunciation model	Which reconstructed or classroom pronunciation standard will govern all authoritative audio?
Text adaptation	How quickly should readings move from authored/adapted Greek to lightly adapted or authentic Xenophon?
Assessment policy	Which activities are practice, which are graded, and what score or retry rule marks lesson completion?
Lesson release	Should progression be sequentially gated, professor-released, date-released, or open by default?
Module I scope	Are Lessons 1–4 the minimum completed set for fall 2026, or should Lessons 5–6 be formal deliverables?
Hosting plans	Which Netlify and Neon account plans are active, and what monthly budget or automatic-upgrade policy should govern production use?
Comparative Plato	Where does a Plato passage materially strengthen the Xenophon curriculum without diluting its organizing principle?
Accessibility review	Who will conduct final keyboard/screen-reader testing, and which assistive technology/browser combinations are required?

Appendix A. Initial Requirements Register

Requirement	Area	Status	Priority
AUTH-01	Accounts and active roles	Implemented	High
NAV-01-08	Role-aware persistent navigation	Partially implemented	High
CUR-01	Unit 0 plus four twelve-lesson modules	Planned	High
LES-01-10	Shared three-page lesson experience	Partially implemented	High
VOC-01	Intentional vocabulary source boundary	Implemented	High
KEY-01-07	Polytonic Greek keyboard	Implemented	High
AUD-01	Word and passage audio with text alternatives	Partially implemented	Medium
PRG-01	Progress, grades, activity, XP, and achievements	Partially implemented	High
PROF-01	Professor roster and analytics	Implemented	High
ADMIN-01	Draft, preview, validate, publish lessons	Partially implemented	High
TECH-01	Netlify, Functions, Neon, and Blobs architecture	Implemented	High
CAP-01	Connection pooling, capacity testing, and quota monitoring	Planned	High
A11Y-01-14	WCAG 2.2 AA target and acceptance tests	Partially implemented	High
SEC-01	Server-side authorization and secret handling	Partially implemented	High
NFR-01	Performance, responsive design, reliability, privacy	Partially implemented	Medium

Appendix B. Source and Traceability Notes

This draft was prepared from the current application repository and the earlier project specification. The following artifacts are the principal implementation references used to distinguish existing behavior from planned scope.

Specification area	Primary source
Project vision and pedagogy	docs/project-spec.md; docs/ideas.md
Modules, lesson sequence, roles, navigation, dashboard	script.js; index.html; lessons.html; module-*.html
Lesson rendering and content shape	lesson.html; lesson.js; lesson-data.js; netlify/functions/lesson-content.mts
Administrative content workflow	netlify/functions/admin-lesson-content.mts; database content tables/migrations
Professor progress view	netlify/functions/professor-dashboard.mts; script.js
Greek keyboard	greek-keyboard.js; course-introduction.html; lesson-introduction.html; dictionary pages
Vocabulary source boundary	netlify/functions/dictionary.mts; netlify/functions/_shared/dictionary-core.mjs; lesson_vocabulary and reading_glosses data paths
Forms and principal parts	forms-data.js; forms.js; forms*.html; principal-parts-data.js; principal-parts*.html/js
Hosting and database architecture	netlify.toml; package.json; netlify/functions/*.mts; db/README.md; db/migrations/*.sql; live read-only database inspection
Provider limits	Neon pricing, compute, and connection-pooling documentation; Netlify pricing and credit-plan documentation; checked August 5, 2026
Accessibility standard	W3C WCAG 2 Overview and WCAG 2.2 Recommendation

Revision Notes for the Next Review

- Confirm the formal independent-study deliverable count for Module I lessons.
- Confirm whether the specification should describe the current GREK 110 J10 course instance or a reusable course platform independent of section and term.
- Review the twelve Module I lesson titles and grammar sequence with Dr. Beck.
- Decide which administrator and professor capabilities must be fully functional by the final demonstration.
- Replace visual-reference figures with updated production screenshots as the interface stabilizes.